Multiple Choice Questions

MCQ on Balanced Diet and Nutritional Requirements

By Biology Ease / May 9, 2025 / 6 min read

Easy Level (1–20)

1. **A balanced diet contains:**
   a) Only proteins
   b) Only carbohydrates
   c) All essential nutrients ✅
   d) Only fats
   Explanation: A balanced diet includes carbohydrates, proteins, fats, vitamins, minerals, water, and fiber.
2. **Which of the following is a macronutrient?**
   a) Iron
   b) Vitamin A
   c) Protein ✅
   d) Zinc
   Explanation: Macronutrients include carbohydrates, proteins, and fats.
3. **Which nutrient is the body's main source of energy?**
   a) Protein
   b) Carbohydrates ✅
   c) Fats
   d) Vitamins
   Explanation: Carbohydrates are the body's primary energy source.

4. **Which food is rich in protein?**
   a) Rice
   b) Eggs ✅
   c) Butter
   d) Sugar
   Explanation: Eggs are a good source of complete protein.
5. **What nutrient helps build and repair body tissues?**
   a) Carbohydrates
   b) Fats
   c) Proteins ✅
   d) Water
   Explanation: Proteins are used for tissue repair and growth.
6. **Which vitamin is obtained from sunlight?**
   a) Vitamin A
   b) Vitamin B12
   c) Vitamin D ✅
   d) Vitamin K
   Explanation: Sunlight helps the body synthesize vitamin D.
7. **Water is essential for:**
   a) Providing calories
   b) Breaking down protein
   c) Hydration and metabolism ✅
   d) Building muscles
   Explanation: Water maintains hydration and helps in all metabolic activities.
8. **Which of the following is a rich source of calcium?**
   a) Banana
   b) Milk ✅
   c) Spinach
   d) Carrot
   Explanation: Milk and dairy products are rich in calcium.
9. **Iron is important for:**
   a) Vision

b) Bone strength

c) Blood production ✅

d) Skin health

Explanation: Iron is essential for hemoglobin in red blood cells.

10. **Which nutrient prevents constipation?**

a) Protein

b) Fiber ✅

c) Fat

d) Salt

Explanation: Fiber adds bulk to stool and promotes bowel movement.

11. **Vitamin C deficiency causes:**

a) Rickets

b) Scurvy ✅

c) Night blindness

d) Anemia

Explanation: Vitamin C deficiency leads to bleeding gums and scurvy.

12. **Which is a good source of vitamin A?**

a) White rice

b) Carrots ✅

c) Bread

d) Cheese

Explanation: Carrots are rich in beta-carotene, a precursor of vitamin A.

13. **Fats are required in a balanced diet because they:**

a) Cause obesity

b) Provide vitamins ✅

c) Weaken bones

d) Build enzymes

Explanation: Fats help in the absorption of fat-soluble vitamins.

14. **Which vitamin helps in blood clotting?**

a) Vitamin A

b) Vitamin B

c) Vitamin C

d) Vitamin K ✅

Explanation: Vitamin K aids in the clotting of blood.

15. **Which mineral is essential for strong bones and teeth?**

a) Iron

b) Potassium

c) Calcium ✅

d) Iodine

Explanation: Calcium is needed for bone strength.

16. **Proteins are made of:**

a) Glucose

b) Amino acids ✅

c) Fatty acids

d) Nucleotides

Explanation: Proteins are built from amino acids.

17. **A deficiency of iodine causes:**

a) Rickets

b) Goiter ✅

c) Scurvy

d) Anemia

Explanation: Iodine is needed for thyroid hormone synthesis.

18. **Which of the following is NOT part of a balanced diet?**

a) Fruits

b) Whole grains

c) Sugary snacks ✅

d) Vegetables

Explanation: Sugary snacks provide empty calories and lack nutrients.

19. **Children need more of which nutrient for growth?**

a) Protein ✅

b) Fat

c) Salt

d) Water

Explanation: Proteins help in tissue development.

20. Which food group gives the body quick energy?

a) Vitamins

b) Proteins

c) Carbohydrates ✅

d) Minerals

Explanation: Carbs are the fastest energy source.

Biological Sciences

🟡 Moderate Level (21–40)

21. Which vitamin is essential for vision?

a) Vitamin A ✅

b) Vitamin D

c) Vitamin B6

d) Vitamin E

Explanation: Vitamin A helps maintain healthy vision.

22. Balanced diet varies based on:

a) Age and activity level ✅

b) Weather

c) Nationality

d) Clothing style

Explanation: Nutrient needs change with age and lifestyle.

23. The nutrient that helps regulate body temperature is:

a) Carbohydrates

b) Water ✅

c) Vitamins

d) Minerals

Explanation: Water regulates body heat through sweating.

24. Which vitamin is an antioxidant?

a) Vitamin D

b) Vitamin K

c) Vitamin E ✅

d) Vitamin B12

Explanation: Vitamin E protects cells from oxidative damage.

25. **Rickets in children is caused by deficiency of:**

a) Iron

b) Vitamin C

c) Vitamin D ✅

d) Vitamin A

Explanation: Vitamin D is required for bone development.

26. **Which nutrient helps in wound healing?**

a) ⚡ Vitamin C ✅

b) ⚡ Vitamin D

c) Vitamin K

d) Vitamin B12

Explanation: Vitamin C supports collagen production and healing.

27. **Good dietary sources of iron include:**

a) Milk and cheese

b) Meat and spinach ✅

c) Fruits and nuts

d) White bread

Explanation: Red meat and leafy greens provide bioavailable iron.

28. **Which ⚡ food provides complete protein?**

a) Wheat

b) Soybeans ✅

c) Maize

d) Potato

Explanation: Soy contains all essential amino acids.

29. **Which of the following is a water-soluble vitamin?**

a) Vitamin A

b) Vitamin D

c) Vitamin C ✅

d) Vitamin E

Explanation: Vitamin C dissolves in water.

30. **Pregnant women need more:**
    a) Protein and iron ✅
    b) Carbs and fat
    c) Sodium
    d) Sugar
    Explanation: Extra nutrients support fetal growth.

31. **Body Mass Index (BMI) helps assess:**
    a) Intelligence
    b) Weight status ✅
    c) Bone strength
    d) Vitamin level
    Explanation: BMI is a weight-to-height ratio.

32. **Folic acid is important for:**
    a) Eye health
    b) Brain development in fetus ✅
    c) Hair growth
    d) Bone density
    Explanation: Folic acid prevents neural tube defects.

33. **Energy needs are highest in which group?**
    a) Toddlers
    b) Adults
    c) Teenagers ✅
    d) Elderly
    Explanation: Teenagers grow rapidly and need more energy.

34. **The main function of roughage (fiber) is to:**
    a) Provide energy
    b) Build tissues
    c) Aid digestion ✅
    d) Build hormones
    Explanation: Fiber helps move food through the gut.

35. **A vegetarian diet may lack:**
    a) Carbohydrates

b) Fiber

c) Vitamin B12 ✅

d) Vitamin C

Explanation: B12 is found mostly in animal products.

36. **Which combination gives all essential amino acids?**

a) Rice and dal ✅

b) Wheat and potato

c) Bread and butter

d) Corn and sugar

Explanation: Cereals and pulses complement amino acid profiles.

37. **Too much salt in the diet may cause:**

a) Anemia

b) High blood pressure ✅

c) Rickets

d) Dehydration

Explanation: Excess sodium can raise blood pressure.

38. **The primary function of vitamins is to:**

a) Build muscle

b) Regulate body functions ✅

c) Provide energy

d) Form bones

Explanation: Vitamins help enzymes and processes.

39. **Which nutrient is not digested but adds bulk to diet?**

a) Fiber ✅

b) Starch

c) Protein

d) Sugar

Explanation: Fiber is not digested but aids in elimination.

40. **A lacto-vegetarian avoids:**

a) Milk

b) Eggs and meat ✅

c) Vegetables

d) Fruits

Explanation: Lacto-vegetarians consume dairy but not eggs/meat.

Vitamins & Supplements

🔴 Hard Level (41–50)

41. **Basal Metabolic Rate (BMR) refers to:**
    a) Maximum calories burned in exercise
    b) Energy needed at rest ✅
    c) Fat burned after food
    d) Energy to digest food
    Explanation: BMR is the resting energy requirement.

42. **Kwashiorkor is caused by deficiency of:**
    a) Vitamin D
    b) Carbohydrates
    c) Protein ✅
    d) Iron
    Explanation: Kwashiorkor results from protein deficiency.

43. **Marasmus is caused by deficiency of:**
    a) Protein only
    b) Carbs only
    c) Overall calories ✅
    d) Vitamin C
    Explanation: Marasmus is due to general undernutrition.

44. **Anemia may result from deficiency of:**
    a) Iodine
    b) Iron ✅
    c) Calcium
    d) Vitamin D
    Explanation: Iron is essential for hemoglobin production.

45. Which vitamin helps calcium absorption?

a) Vitamin B12

b) Vitamin A

c) Vitamin D ✅

d) Vitamin K

Explanation: Vitamin D promotes calcium absorption in intestines.

46. Which is NOT a function of water in the body?

a) Regulates temperature

b) Aids digestion

c) Builds muscles ✅

d) Carries nutrients

Explanation: Water doesn't build tissue, but supports processes.

47. Balanced diet for elderly should be:

a) High calorie

b) High protein

c) Low in fat and sodium ✅

d) High in sugar

Explanation: Reducing fat and sodium helps prevent chronic diseases.

48. Energy value of 1g of carbohydrate is:

a) 2 kcal

b) 4 kcal ✅

c) 6 kcal

d) 9 kcal

Explanation: 1 gram of carbohydrate provides 4 kcal.

49. Zinc is required for:

a) Vision

b) Wound healing ✅

c) Blood clotting

d) Energy production

Explanation: Zinc supports healing and immunity.

50. Night blindness is caused by deficiency of:

a) ⚡ Vitamin A ✅

b) Vitamin C

c) Vitamin D

d) Vitamin E

Explanation: Vitamin A is vital for vision, especially in low light.

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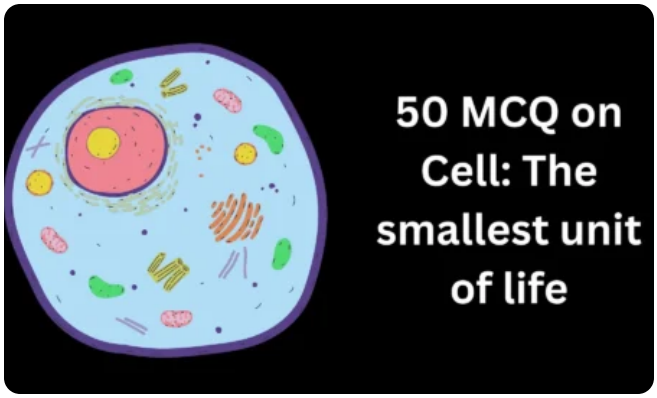

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